

UC-1: Manage Items & Inventory

Name: Manage Items & Inventory

Actors: Cachier, Store Manager

Goals:

- Create, edit, and deactivate items with barcodes, units, and prices.
- Maintain accurate inventory quantities.

Preconditions:

- Actor is authenticated and has inventory permissions.
- Store/tenant is already configured.

Summary: Actor manages the master data for items (products) including barcodes, units, prices, and initial stock.

Related use cases:

- UC-2 Monitor Inventory Levels & Alerts
- UC-3 Perform Inventory Adjustment/Reconciliation

Steps:

1. Actor navigates to the “Items & Inventory” screen.
2. Actor selects “Add New Item”.
3. System prompts for item details (name, barcode(s), unit(s), category, tax group, price).
4. Actor fills in required fields and saves.
5. System validates data and creates the item record.
6. Actor optionally enters opening quantity or imports from a file.
7. System updates inventory for the store and logs the action in the audit log.

Postconditions:

- New item(s) are stored and visible in searches and POS.
- Inventory levels are updated accordingly.

UC-2: Monitor Inventory Levels & Alerts

Name: Monitor Inventory Levels & Alerts

Actors: Store Manager, Cachier

Goals: Detect low-stock items before they run out.

Preconditions: Items exist in the system with configured minimum levels.

Summary: Actor views inventory and receives alerts when stock drops below thresholds.

Related use cases:

- UC-1 Manage Items & Inventory
- UC-3 Perform Inventory Adjustment/Reconciliation

Steps:

1. System periodically evaluates stocks against configured thresholds.

2. For items below threshold, system generates alert entries.
3. Actor opens “Inventory Alerts” screen or dashboard.
4. System displays items with low stock and related info.
5. Actor may export the list or start a purchase process.

Postconditions: Low-stock items are identified and visible to responsible users.

UC-3: Perform Inventory Adjustment/Reconciliation

Name: Perform Inventory Adjustment/Reconciliation

Actors: Cashier, Accountant

Goals: Align system inventory with physical stock counts and keep traceable audit.

Preconditions: Items exist with current stock balances.

Summary: Actor records stock corrections based on cycle counts or full inventory checks.

Related use cases:

- UC-1 Manage Items & Inventory
- UC-2 Monitor Inventory Levels & Alerts

Steps:

1. Actor opens “Inventory Reconciliation / Adjustments” screen.
2. Actor selects store, date, and item(s) to reconcile.
3. Actor enters physical counted quantity.
4. System calculates difference and shows adjustment (gain/loss).
5. Actor reviews and confirms adjustments.
6. System posts inventory adjustment entry and logs details (who/when/old vs new quantity).

Postconditions:

- Inventory quantities are updated.
- Audit record is stored for each adjustment.

UC-4: Manage Price Lists & Offers

Name: Manage Price Lists & Offers

Actors: Cashier

Goals: Define multiple price lists and promotional offers.

Preconditions: Items exist in the system.

Summary: Manager creates or updates price lists and offers that affect selling prices in POS and e-commerce.

Related use cases:

- UC-1 Manage Items & Inventory
- UC-5 Process POS Sale

Steps:

1. Actor opens “Price Lists / Offers” module.
2. Actor creates a new price list or edits an existing one.
3. Actor selects items and sets prices or discounts (percentage/amount).
4. Actor optionally defines validity period (start/end date).
5. System validates rules and saves the price list/offer.
6. System applies active prices to POS and APIs at runtime.

Postconditions: Active price lists/offers are applied to sales channels.

UC-5: Process POS Sale (Online Mode)

Name: Process POS Sale (Online Mode)

Actors: Cashier, End Customer

Goals: Complete a sales transaction quickly and issue a tax invoice/receipt.

Preconditions:

- POS is connected to the server (online).
- Items, prices, taxes, and payment methods are configured.

Summary: Cashier scans items, applies discounts/taxes, selects payment, and issues receipt.

Related use cases:

- UC-6 Process POS Sale in Offline Mode
- UC-7 Issue Sales Invoice/Return
- UC-13 Record Treasury Receipt/Payment

Steps:

1. Cashier logs in to POS.
2. Cashier starts a new sale.
3. Cashier scans item barcodes or searches for items.
4. System displays items, prices, taxes, and running total.
5. Cashier adjusts quantities and applies valid discounts if needed.
6. Cashier selects payment method (cash, card, etc.).
7. System validates payment and records sale.
8. System reduces inventory quantities accordingly.
9. System prints receipt through thermal printer and optionally sends e-receipt.

Postconditions:

- Sale is stored and reflected in inventory and financial records.
- Receipt/tax invoice is generated.

UC-6: Process POS Sale in Offline Mode with Sync

Name: Process POS Sale in Offline Mode with Sync

Actors: Cashier

Goals: Allow selling even when internet or connection to server is down, then sync later.

Preconditions:

- POS client is installed with offline capability.
- POS has a cached snapshot of items and prices.

Summary: Cashier processes sales locally; when connection is restored, system synchronizes transactions.

Related use cases:

- UC-5 Process POS Sale (Online Mode)
- UC-15 Synchronize Offline Transactions

Steps:

1. System detects loss of connection and switches POS to offline mode.
2. Cashier starts a new sale as usual.
3. POS uses local copy of items and prices to process the sale.
4. Cashier completes payment and prints receipt locally.
5. When connection is restored, POS sends queued transactions to the server.
6. Server validates and posts each transaction, updating inventory and financial records.
7. System marks synced sales as confirmed.

Postconditions: Offline sales are stored locally then synchronized to the central server.

UC-7: Issue Sales Invoice / Return

Name: Issue Sales Invoice / Return

Actors: Accountant, Cashier

Goals: Create invoices and returns with proper tax and accounting entries.

Preconditions:

- Customer and items exist (for invoice).
- Original invoice exists (for return).

Summary: Actor issues sales invoices or returns for POS or back-office sales.

Related use cases:

- UC-5 Process POS Sale
- UC-9 Manage Customer Credit Profile

Steps:

1. Actor opens “Sales Invoices / Returns” module.
2. Actor selects operation type (invoice or return).
3. Actor selects customer (if applicable) and store.
4. Actor adds items (or selects original invoice for return).
5. System calculates totals, taxes, and any discounts.
6. Actor selects payment method or credit terms.
7. Actor confirms and posts the invoice/return.

8. System updates inventory and financial records.

Postconditions:

- Invoice/return is recorded with unique number.
- Stock and customer balance are updated.

UC-8: Manage Customers & Credit (Ajel)

Name: Manage Customers & Credit (Ajel)

Actors: Accountant, Store Manager

Goals: Maintain customer files with credit limits, payment terms, and schedules.

Preconditions: Actor is authenticated with customer management permissions.

Summary: Actor creates or updates customer records including allowed credit and terms.

Related use cases:

- UC-9 Enforce Credit Limit at Sale
- UC-10 Generate Debt Aging Report

Steps:

1. Actor opens “Customers / Credit Profiles” screen.
2. Actor creates a new customer or edits an existing one.
3. Actor enters general data (name, contact, ID, address).
4. Actor sets credit limit, payment term, and allowed grace period.
5. System validates inputs and saves the profile.
6. System links future invoices to this credit configuration.

Postconditions: Customer credit data is available to sales and reporting modules.

UC-9: Enforce Credit Limit at Sale

Name: Enforce Credit Limit at Sale

Actors: Cashier, Accountant, Store Manager

Goals: Prevent exceeding customer credit limit without authorization.

Preconditions:

- Customer has a defined credit limit and outstanding balance.
- Sale is being processed with a credit term.

Summary: During a credit sale, system checks credit limit and blocks or requests approval if exceeded.

Related use cases:

- UC-5 Process POS Sale.
- UC-8 Manage Customers & Credit.

Steps:

1. Cashier creates a sale for a credit customer.
2. System calculates new outstanding balance if this invoice is approved.
3. System compares projected balance with customer's credit limit.
4. If within limit, system continues normally.
5. If exceeding limit, system warns cashier and blocks posting.
6. Cashier may request supervisor/manager approval.
7. Supervisor (Store Manager/Accountant) logs in and overrides if allowed.
8. System records decision in audit log.

Postconditions: Either sale is posted with proper authorization or cancelled/adjusted.

UC-10: Generate Debt Aging Report

Name: Generate Debt Aging Report

Actors: Accountant, Store Manager

Goals: Analyze customer debts by aging buckets (e.g., 0–30, 31–60 days).

Preconditions: There are posted invoices and payments for customers.

Summary: Actor generates a report showing unpaid balances grouped by age.

Related use cases:

- UC-8 Manage Customers & Credit
- UC-9 Enforce Credit Limit at Sale

Steps:

1. Actor opens "Receivables / Debt Aging" report.
2. Actor selects date, store(s), and optional filters (customer group, etc.).
3. System calculates unpaid balances and days past due.
4. System groups balances into aging buckets and displays the report.
5. Actor exports or prints the report if needed.

Postconditions:

- Debt aging report is generated and can be saved/exported.

UC-11: Create Purchase Order

Name: Create Purchase Order

Actors: Inventory Clerk, Store Manager

Goals: Create purchase orders to suppliers based on stock needs.

Preconditions: Suppliers exist in the system.

Summary: Actor creates a purchase order to replenish inventory.

Related use cases: UC-12 Receive Supplier Invoice & Update Stock

Steps:

1. Actor opens “Purchasing / Purchase Orders” module.
2. Actor selects supplier and store.
3. Actor adds items, quantities, and expected prices.
4. System calculates PO total.
5. Actor reviews and saves or submits for approval (if workflow exists).
6. System assigns PO number and status (e.g., Open).

Postconditions: Purchase order is recorded and can be referenced upon receipt.

UC-12: Receive Supplier Invoice & Update Stock

Name: Receive Supplier Invoice & Update Stock

Actors: Inventory Clerk, Accountant

Goals: Record goods received and supplier invoice; adjust quantities and cost.

Preconditions:

- Supplier exists.
- Optionally, a related purchase order exists.

Summary: Actor records receipt of items and links it to supplier invoice.

Related use cases:

- UC-11 Create Purchase Order
- UC-3 Perform Inventory Adjustment/Reconciliation

Steps:

1. Actor opens “Supplier Invoices / Goods Receipt” screen.
2. Actor selects supplier and optionally references an existing PO.
3. System loads PO lines (if any).
4. Actor enters actual received quantities, prices, and invoice details.
5. System calculates total and tax.
6. Actor confirms posting.
7. System updates inventory quantities and item cost.
8. System creates payable entry for the supplier.

Postconditions: Stock and supplier balance are updated.

UC-13: Record Treasury Receipt/Payment/Transfer

Name: Record Treasury Receipt/Payment/Transfer

Actors: Accountant, Cashier

Goals: Track cash movements (in/out/transfer) across cash boxes and banks.

Preconditions: Cash accounts and bank accounts are defined.

Summary: Actor records cash receipts, disbursements, and internal transfers.

Related use cases:

- UC-5 Process POS Sale
- UC-14 Perform Bank Reconciliation

Steps:

1. Actor opens “Treasury / Cash Operations” module.
2. Actor selects operation type (receipt, payment, transfer).
3. Actor selects source and destination accounts where applicable.
4. Actor enters amount, currency, and reference.
5. System validates balances and saves transaction.
6. System updates cash/bank balances and general ledger (if applicable).

Postconditions: Treasury transaction is recorded and affects account balances.

UC-14: Perform Bank Reconciliation & Import Statements

Name: Perform Bank Reconciliation & Import Statements

Actors: Accountant

Goals: Reconcile system bank balance with bank statement.

Preconditions:

- Bank account exists and has transactions.
- Bank statement file is available or data can be entered manually.

Summary: Actor imports or enters bank statement and matches it with system records.

Related use cases: UC-13 Record Treasury Receipt/Payment/Transfer

Steps:

1. Actor opens “Bank Reconciliation” module.
2. Actor selects bank account and period.
3. Actor imports bank statement file or enters data manually.
4. System lists unmatched bank transactions and system transactions.
5. Actor matches transactions (one-to-one or many-to-one).
6. Actor confirms reconciliation.
7. System records reconciliation status and any adjustments.

Postconditions: Bank account is reconciled for the selected period.

UC-15: View Sales & Margin Reports

Name: View Sales & Margin Reports

Actors: Store Manager, Owner, Accountant

Goals: Analyze sales volume and profit margins over time.

Preconditions: Sales invoices/receipts exist.

Summary: Actor generates reports and dashboards for sales and margins.

Related use cases:

- UC-5 Process POS Sale
- UC-7 Issue Sales Invoice/Return

Steps:

1. Actor opens “Reporting / Sales & Margin” section.
2. Actor selects date range, store(s), and filters.
3. System aggregates sales data and calculates margins.
4. System displays charts/tables on dashboard.
5. Actor drills down into details if needed or exports report.

Postconditions: Sales reports are available for decision-making.

UC-16: View Inventory Movement Report

Name: View Inventory Movement Report

Actors: Inventory Clerk, Store Manager

Goals: Understand stock movement (in, out, adjustments) per item.

Preconditions: Inventory transactions exist (purchases, sales, adjustments).

Summary: Actor views item movement and current stock.

Related use cases:

- UC-3 Perform Inventory Adjustment
- UC-12 Receive Supplier Invoice
- UC-5 Process POS Sale

Steps:

1. Actor opens “Inventory Movement / Stock Report”.
2. Actor selects item(s), store(s), and date range.
3. System retrieves all relevant inventory transactions.
4. System displays movement (in/out/balance) chronologically.
5. Actor exports or prints as needed.

Postconditions: Inventory movement report is generated.

UC-17: Manage Users & Roles (RBAC)

Name: Manage Users & Roles (RBAC)

Actors: System Administrator, Store Owner/Manager

Goals: Define user accounts and assign precise roles/permissions.

Preconditions:

- Tenant/store exists.
- Admin user is logged in.

Summary: Admin creates users and assigns roles controlling access rights.

Related use cases:

- UC-18 View Audit Log
- All other use cases (permissions check).

Steps:

1. Admin opens “Users & Roles” module.
2. Admin creates or edits roles with specific permissions.
3. Admin creates or edits user accounts (login, name, contact).
4. Admin assigns one or more roles to each user.
5. System saves roles and user assignments.
6. System enforces these permissions on every operation.

Postconditions: User and role configuration is saved and applied across the system.

UC-18: View Audit Log

Name: View Audit Log

Actors: System Administrator, Store Owner/Manager

Goals: Track critical actions performed by users for accountability and security.

Preconditions: Audit logging is enabled.

Summary: Actor reviews audit trails for sensitive operations.

Related use cases:

- UC-17 Manage Users & Roles
- UC-3, UC-5, UC-7, UC-9, UC-12 (all log actions).

Steps:

1. Actor opens “Audit Log” screen.
2. Actor filters by date, user, module, or action type.
3. System retrieves matching log entries.
4. System shows who did what, when, and on which records.
5. Actor exports logs if required for compliance.

Postconditions: Audit information is accessible for review and investigation.

UC-19: Expose Items & Stock via E-commerce API

Name: Expose Items & Stock via E-commerce API

Actors: Customer App (External System), API Client

Goals: Allow external web/mobile apps to view items, prices, and availability.

Preconditions:

- API credentials are configured.
- Items and inventory exist.

Summary: External clients call API endpoints to retrieve items and stock.

Related use cases: UC-20 Create Customer Order via API

Steps:

1. External client authenticates via OAuth2/OIDC.
2. Client calls “List Items” or “Get Item Details” API.
3. System validates token and permissions.
4. System returns item data with prices and available stock.

Postconditions: Up-to-date catalog data is available for customer-facing apps.

UC-20: Create Customer Order via API

Name: Create Customer Order via API

Actors: Customer App (External System), End Customer

Goals: Place customer orders from e-commerce front-end into SmartMarket.

Preconditions:

- Items and customers (or guest rules) exist.
- Integration is configured.

Summary: External app sends order details; system creates corresponding order/transaction.

Related use cases:

- UC-19 Expose Items & Stock via API
- UC-21 Manage Delivery & Shipment
- UC-22 Process Online Payment

Steps:

1. End customer selects items in web/mobile app.
2. App sends “Create Order” request to SmartMarket API.
3. System validates customer, items, prices, and stock.
4. System creates internal order with status (e.g., New/Pending).
5. System returns order ID and status to external app.

Postconditions: Order is recorded and ready for fulfillment, payment, and delivery.

UC-21: Manage Delivery & Shipment

Name: Manage Delivery & Shipment

Actors: Store Staff (Fulfillment), Delivery Provider (External System)

Goals: Arrange shipment and track delivery status for orders.

Preconditions:

- Order exists and is ready to ship.
- Delivery integration is configured.

Summary: System integrates with delivery providers to create shipments and receive status updates.

Related use cases:

- UC-20 Create Customer Order via API
- UC-22 Process Online Payment

Steps:

1. Actor opens order fulfillment screen.
2. Actor selects order and chooses a delivery provider.
3. System sends shipment creation request to provider API with order details.
4. Provider returns tracking number and estimated delivery info.
5. System stores tracking number and updates order status (e.g., Shipped).
6. Provider sends webhooks or status updates (In Transit, Delivered, Failed).
7. System updates order status accordingly.

Postconditions: Shipment is created and tracked within SmartMarket.

UC-22: Process Online Payment / Cash on Delivery

Name: Process Online Payment / Cash on Delivery

Actors: End Customer, Payment Gateway (External System), Cashier

Goals: Collect payment via multiple gateways or cash on delivery.

Preconditions:

- Order or sale exists.
- Payment gateway integration and methods are configured.

Summary: Customer pays online or in cash; system confirms or updates payment status.

Related use cases:

- UC-5 Process POS Sale
- UC-20 Create Customer Order via API

Steps (Online Payment):

1. System redirects customer to selected payment gateway or displays embedded form.
2. Customer enters payment details and confirms.
3. Gateway processes payment and returns success/failure callback.
4. System updates order/payment status and logs transaction reference.

Steps (Cash on Delivery):

5. Delivery agent collects cash at delivery.
6. Cashier or accountant records COD receipt in SmartMarket.

Postconditions: Payment status is stored and linked to the order/invoice.

UC-23: Configure Store Settings & Localization

Name: Configure Store Settings & Localization

Actors: Store Owner/Manager, System Administrator

Goals: Configure VAT/tax, currency, language, and formats per store.

Preconditions:

- Store/tenant is created.
- Actor has admin privileges.

Summary: Actor configures localization and business rules affecting the whole tenant.

Related use cases:

- UC-5 Process POS Sale
- UC-15 View Sales & Margin Reports

Steps:

1. Actor opens “Settings” module.
2. Actor configures tax rates, tax IDs, and invoice formats.
3. Actor selects default language (Arabic/English) and layout (RTL/LTR).
4. Actor sets base currency and number/date formats.
5. System validates and applies settings to all relevant modules.

Postconditions: Store operates with correct tax, language, and formatting settings.